



Friction

Quick Revision Notes · Theory & Standard Results · JEE Main / Advanced · NEET

■ Nature & types

Component of the contact force **parallel** to the surfaces (N = perpendicular component). Arises from irregularities & adhesion at the actual points of contact.

Static

no relative sliding

Limiting

on the verge of slipping

Kinetic

during sliding

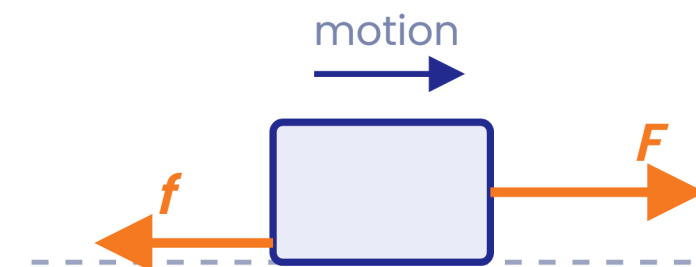
Rolling

much smaller than kinetic

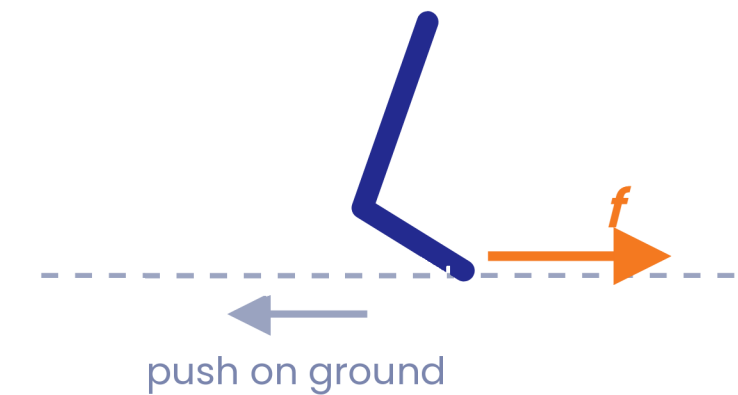
■ Direction rule

KEY

Friction opposes **relative slipping or its tendency** — not necessarily the body's motion.



Pulled block → friction
backward



Walking foot → friction **forward**

■ Laws of friction

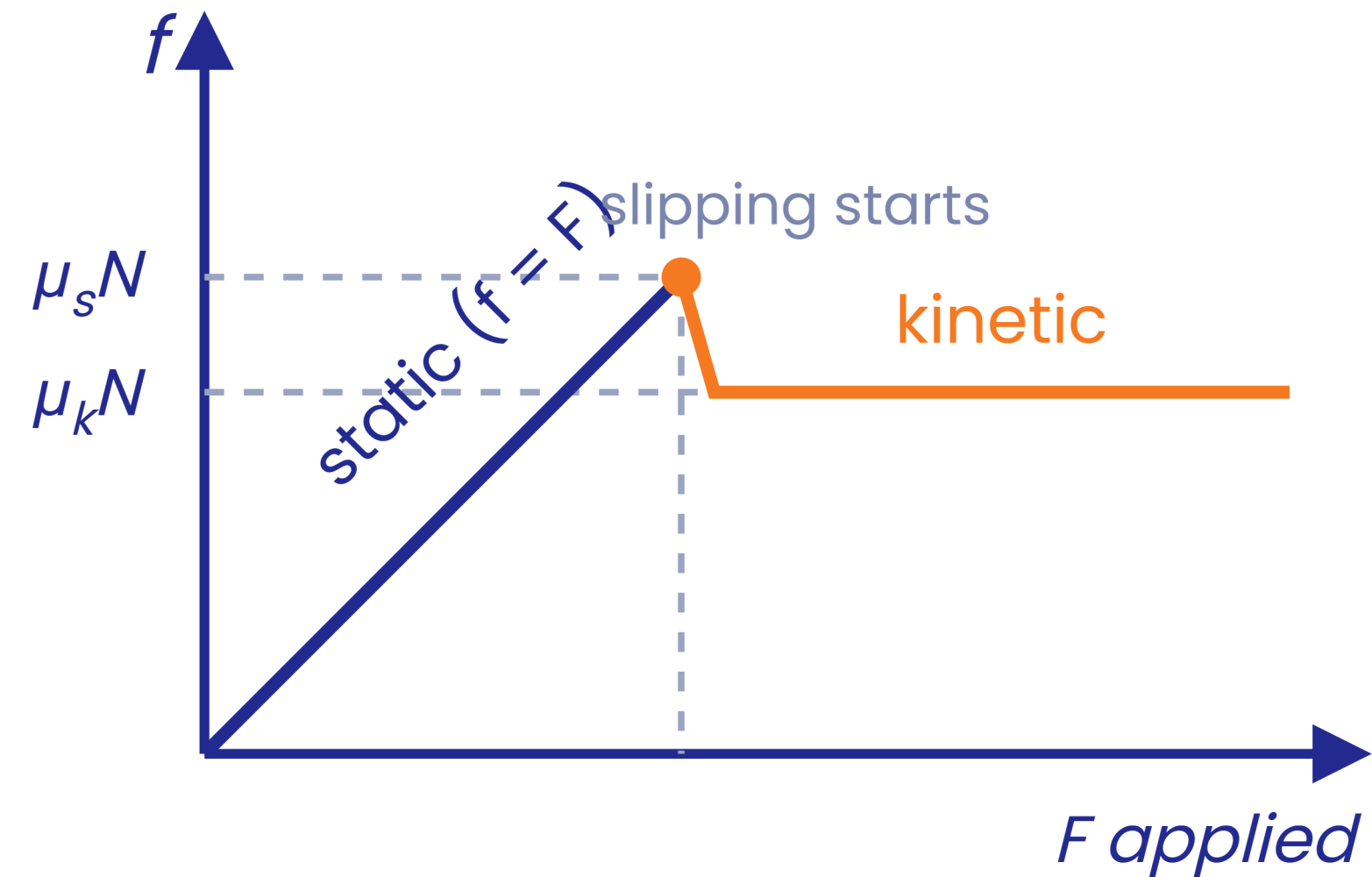
$$f_s \leq \mu_s N \text{ (static – self-adjusting)}$$

$$f_L = \mu_s N \text{ (limiting value)}$$

KEY

$$f_k = \mu_k N, \quad \mu_k < \mu_s$$

■ Friction vs applied force



Independent of the **apparent** area of contact

Depends on the nature of the surfaces and on N

f_k is nearly independent of sliding speed

■ Angle of friction λ

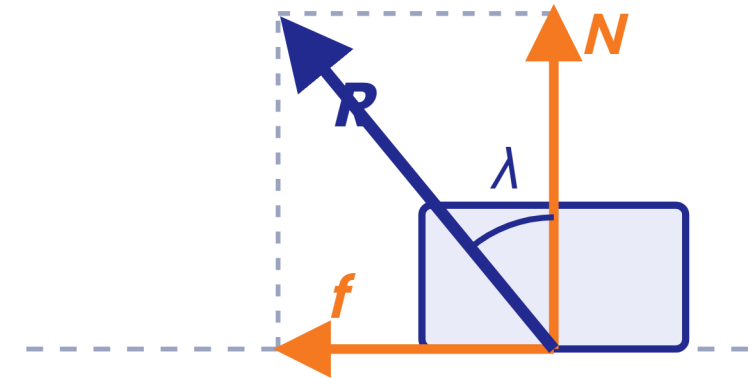
Angle between the **total contact force** R and the **normal** N , at the point of slipping.

KEY

$$\tan \lambda = \mu_s$$

Note: λ is a property of the surface pair — same $\mu_s \Rightarrow$ same λ , whatever the weight.

■ Net contact force



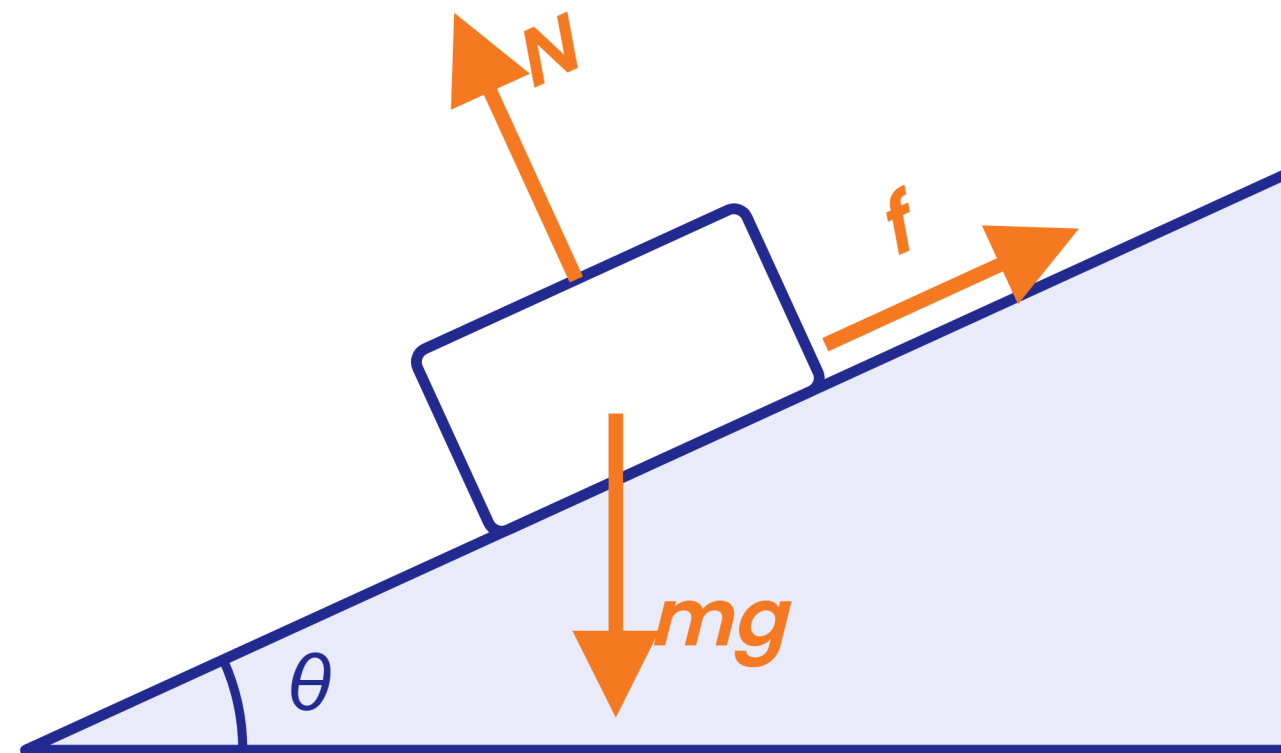
$$R = \sqrt{N^2 + f^2}$$

$$N \leq R \leq N\sqrt{1 + \mu_s^2}$$

from N (no friction) to $N\sqrt{1 + \mu_s^2}$ (limiting)

■ Angle of repose θ_r

Maximum incline angle for which the block stays at rest.



KEY

$$\tan \theta_r = \mu_s, \quad \theta_r = \lambda$$

■ At rest ($\theta \leq \theta_r$)

$$\text{at rest} \Leftrightarrow \tan \theta \leq \mu_s$$

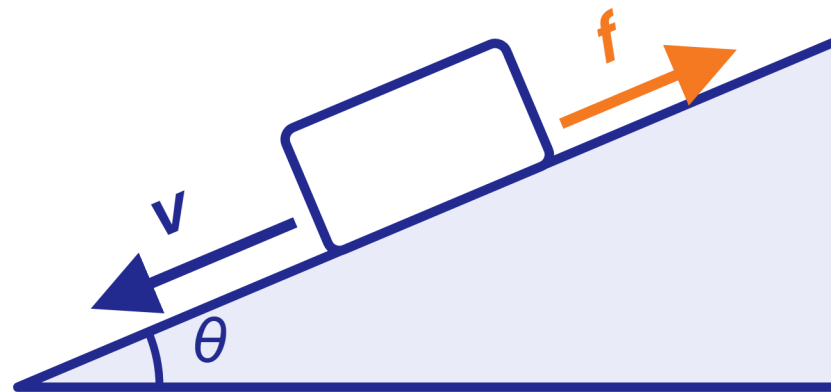
Friction is **self-adjusting** — solve from equilibrium:

$$f = mg \sin \theta \quad (\text{NOT } \mu mg \cos \theta)$$

$$N = mg \cos \theta$$

Trap: use $f = \mu N$ only at the verge of slipping or during sliding.

Sliding down



$$a = g(\sin \theta - \mu_k \cos \theta)$$

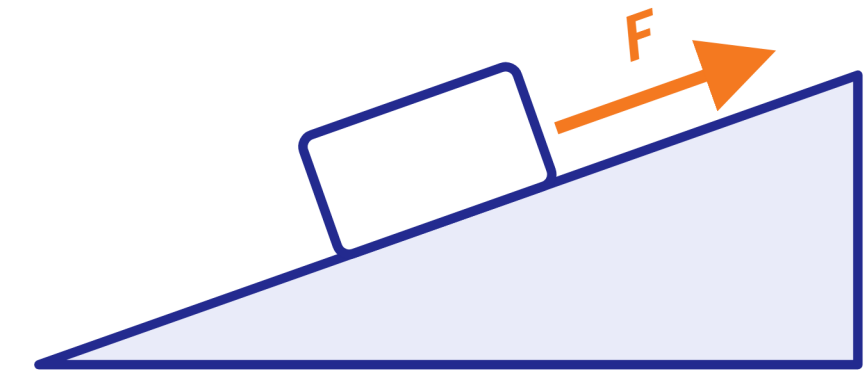
Projected up



$$a = g(\sin \theta + \mu_k \cos \theta)$$

Note: slides back only if $\tan \theta > \mu_s$

Force along the incline



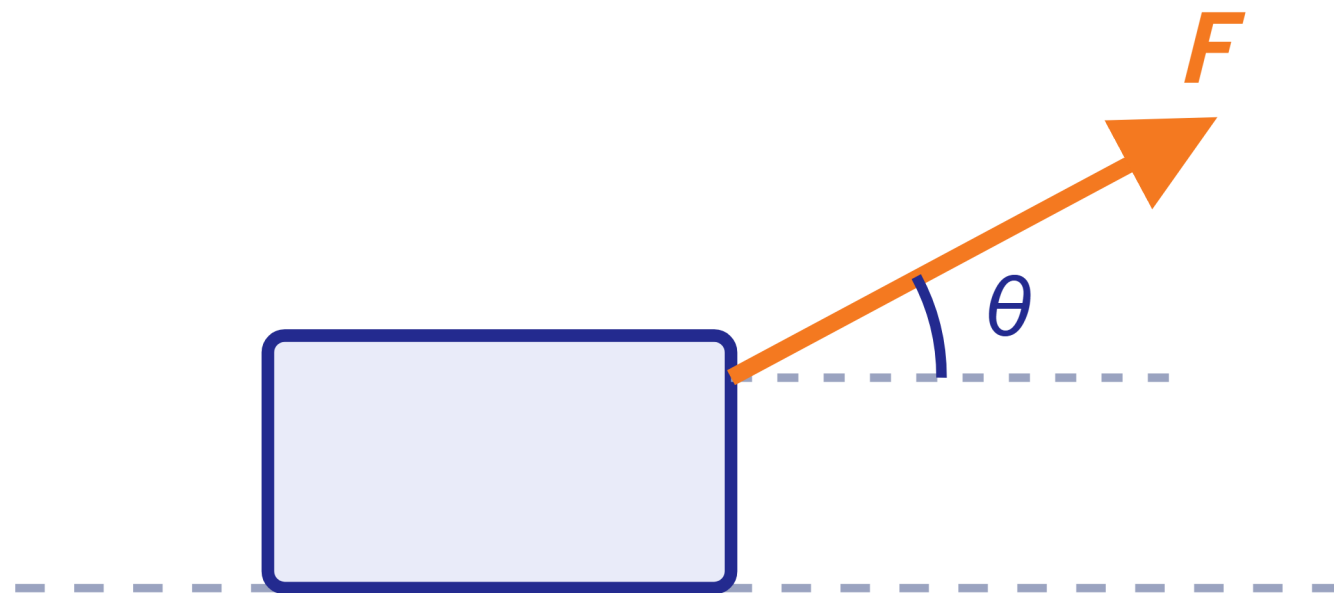
push up at constant velocity

$$F = mg(\sin \theta + \mu_k \cos \theta)$$

just prevent sliding down

$$F = mg(\sin \theta - \mu_s \cos \theta)$$

■ Pull at angle θ above horizontal



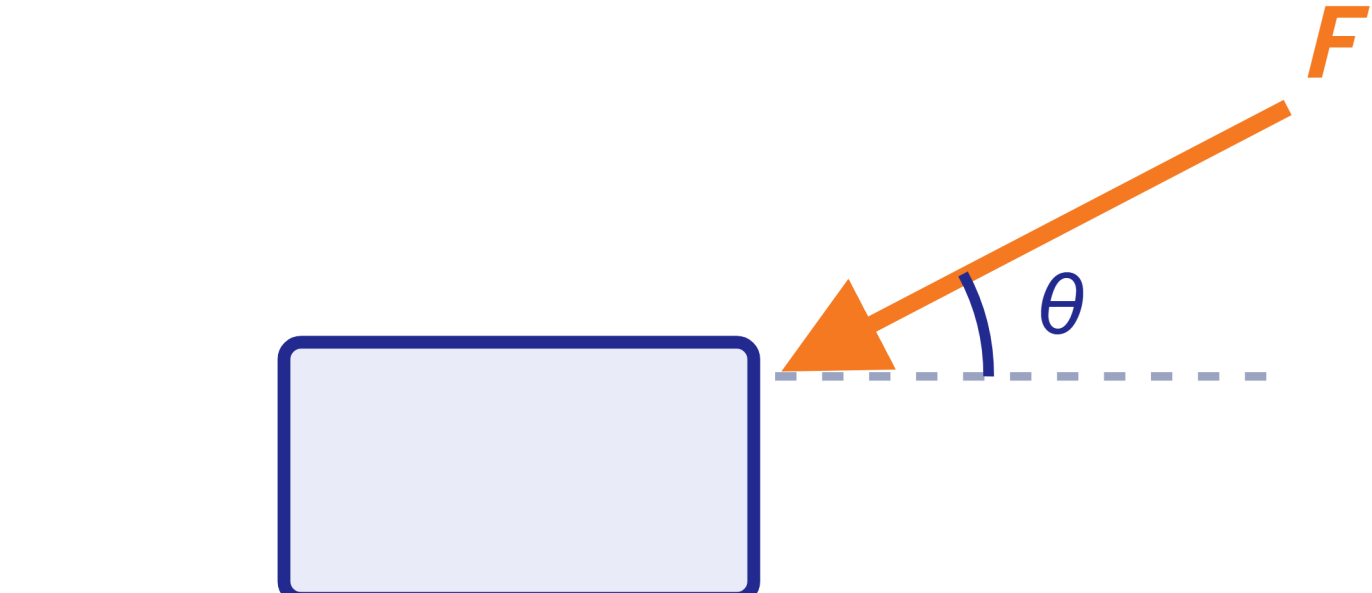
$$N = mg - F \sin \theta$$

$$F = \frac{\mu mg}{\cos \theta + \mu \sin \theta}$$

KEY

$$\text{at } \theta = \lambda: F_{\min} = \frac{\mu mg}{\sqrt{1 + \mu^2}} = mg \sin \lambda$$

■ Push at angle θ below horizontal

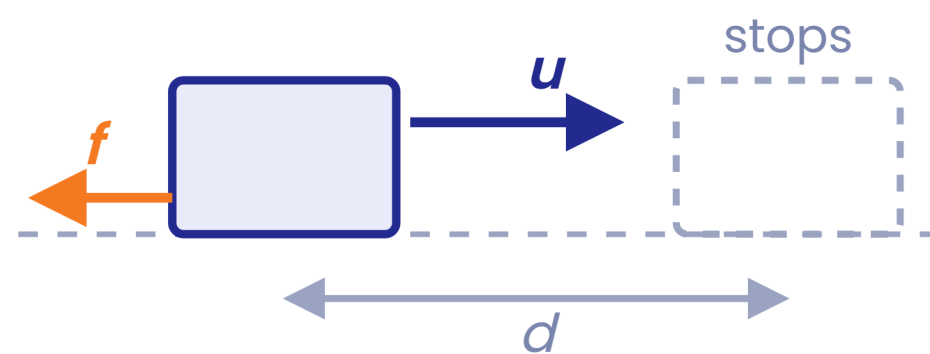


$$N = mg + F \sin \theta$$

$$F = \frac{\mu mg}{\cos \theta - \mu \sin \theta}$$

Note: always larger than the pull $\Rightarrow$ **pulling is easier than pushing.**

■ Sliding to rest

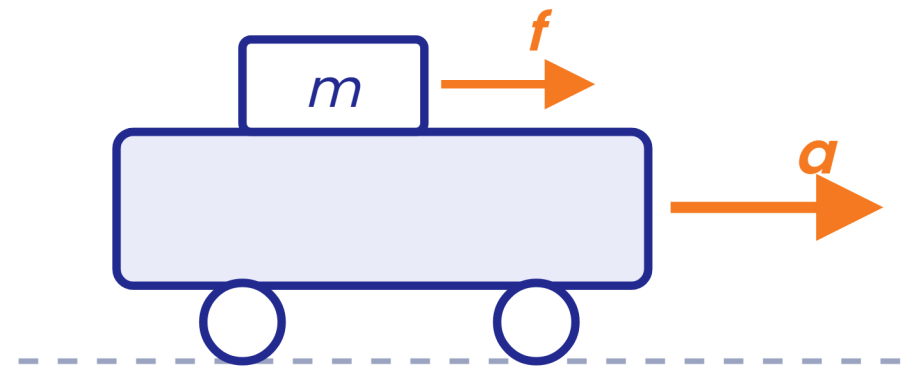


retardation $a = \mu_k g$

KEY

$$d = \frac{u^2}{2\mu_k g}, \quad t = \frac{u}{\mu_k g}$$

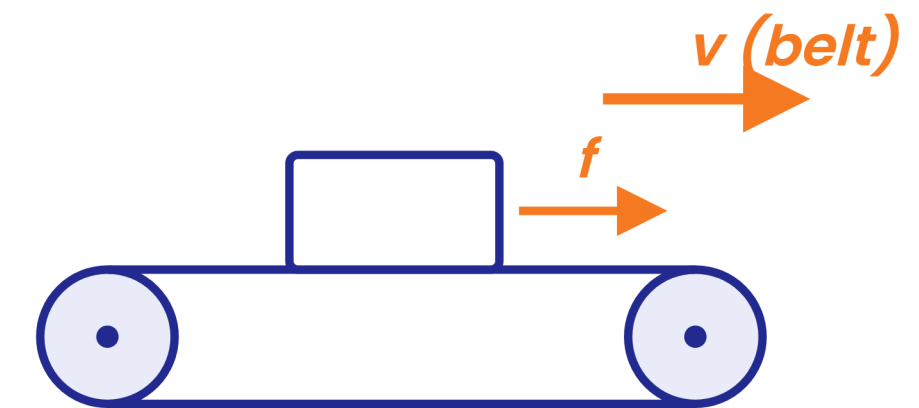
■ Block on a truck floor



no slipping $\Leftrightarrow a_{truck} \leq \mu_s g$

Note: faster truck $\Rightarrow$ block slips back;
kinetic friction gives it $a = \mu_k g$ forward.

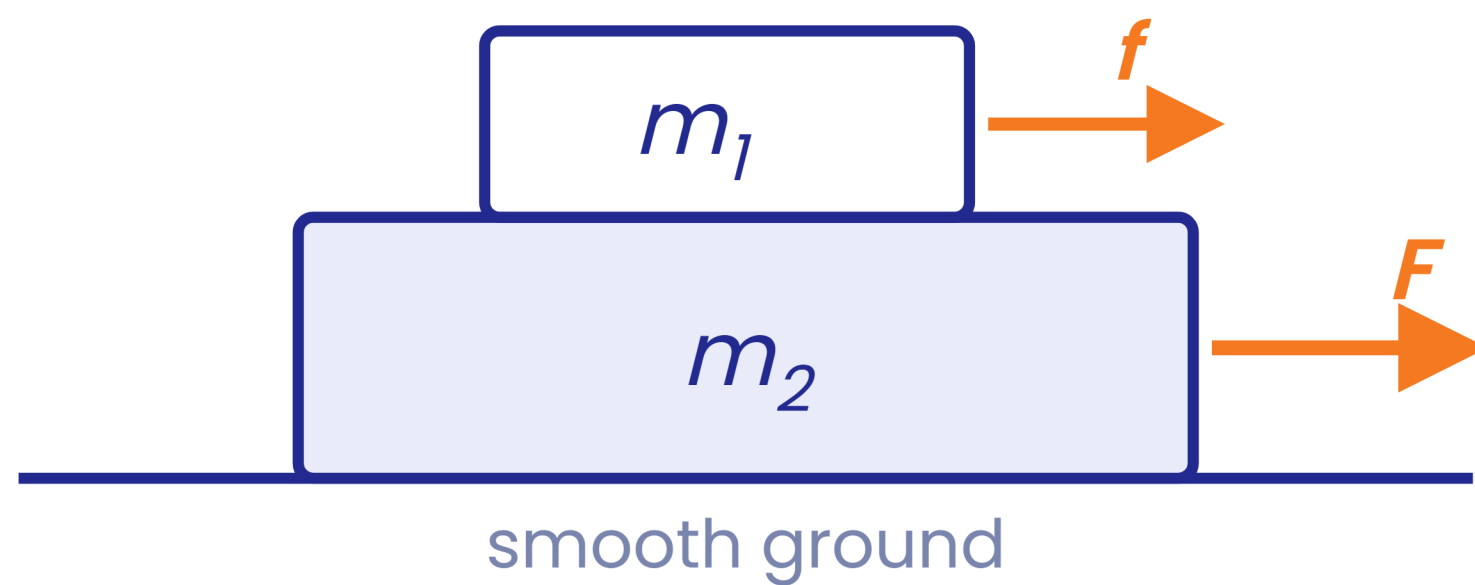
■ Conveyor belt



accelerates at $\mu_k g$ until $v_{block} = v_{belt}$

then moves with the belt under **static**
friction

Standard results (F on m_2 , ground smooth)



KEY

friction-driven block: $a_{max} = \mu_s g$

slipping starts when $F > \mu_s(m_1 + m_2)g$

after slipping: $a_1 = \mu_k g$, $a_2 = \frac{F - \mu_k m_1 g}{m_2}$

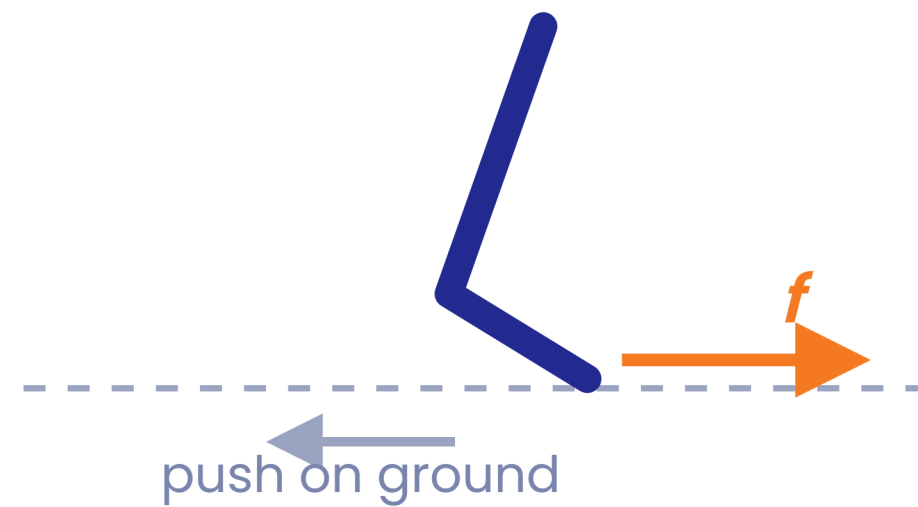
Method — together or slip?

- 1 Assume they move together:

$$a = \frac{F}{m_1 + m_2}$$

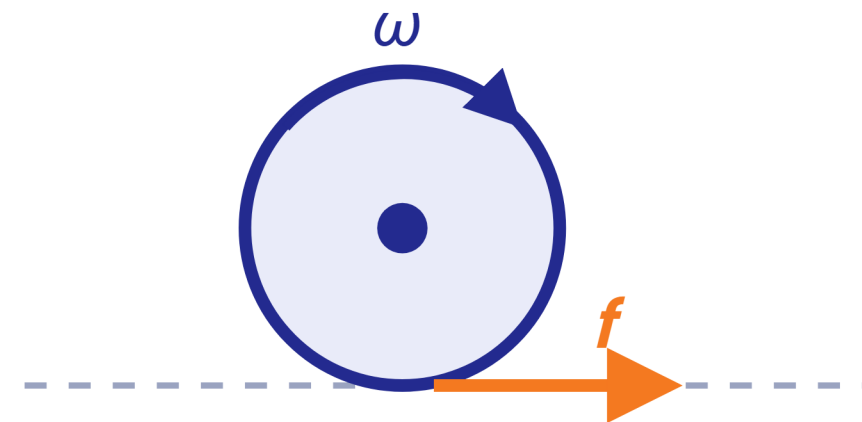
- 2 Find the friction **required** on the friction-driven block.
- 3 Compare with the limiting value $\mu_s N$.
- 4 If it exceeds the limit $\Rightarrow$ they **slip**: use kinetic friction and write separate equations for each block.

Walking



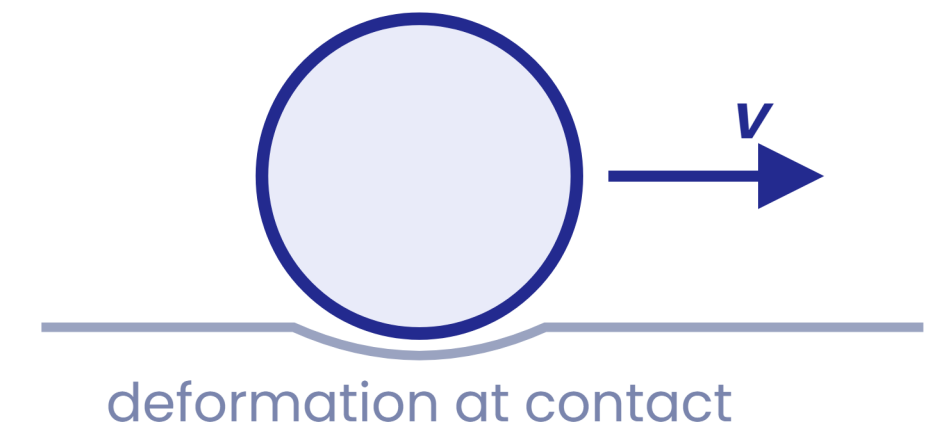
Foot pushes the ground **backward**; forward **static** friction on the foot drives the walker.

Vehicles



Driving wheels get **forward** friction from the road; while braking, friction acts **backward**. No friction $\Rightarrow$ no walking, driving, or braking.

Rolling friction



Caused by deformation at the contact; **much smaller** than sliding friction — why wheels & ball bearings reduce resistance.

- 1 $f = \mu N$ holds **only** at limiting or during sliding — static friction is self-adjusting; solve for it from equilibrium.
- 2 Use μ_s at the **verge of slipping**, μ_k **during sliding**.
- 3 Friction **can act along the motion** — walking, or the upper block carried by the lower one.
- 4 Always check “**move together vs slip**” before writing equations in multi-body problems.
- 5 $N \neq mg$ when applied forces have vertical components, or on an incline — solve for N first, then friction.
- 6 Friction opposes **relative** motion between the surfaces — not the motion of the body itself.